

EMILY LIANG

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Education

The University of Texas at Austin

May 2028 (Expected)

B.S. in Computer Science and Double major in Mathematics + Minor in Business

Austin, TX

Honors: Turing Scholars Honors Program **GPA:** 3.87

Relevant Coursework

- Algorithms (H)
- Discrete Math (H)
- Computer Vision
- Differential Equations
- Computer Architecture (H)
- Data Structures (H)
- Software Engineering
- Linear Algebra
- Operating Systems (H)
- Systems of ML (H)
- Stochastic Processes
- Probability

Experience

Software Engineering Intern

Incoming summer 2026

Fidelity

Fort Worth, TX

Longhorn Racing Internal Combustion

Aug. 2024 – Present

Simulation and Validation — Validation Lead

Austin, TX

- Collaborated on LapSim Kinematics Model (vehicular simulation) to drive 10+ engineering design choices.
- Implement the Magic Formula Tire Model for UT Austin's Formula SAE racing team.

Software Engineering Intern

Sept. 2023 – May 2024

Lockheed Martin Aeronautics

Fort Worth, TX

- Developed a version compatibility lookup tool, reducing software mismatch by 70% and streamlining coordination.
- Collaborated with the Air Force Research Laboratory, L3 Harris, and Raytheon on research efforts.

Technology Project Manager

Sept. 2023 – Sep. 2024

Keller Independent School District

Keller, TX

- Led a team of 10 to support district-wide technology deployment for 35,000 students and 4,000 faculty members.
- Delivered training to 100+ educators on EdTech platforms, reducing software onboarding issues by 50% and ensuring seamless technology adoption across 40+ campuses.

Projects

Foreign Currency Exchange Purchasing Platform | *TypeScript, Node.js, SQL*

Aug. 2025 - Present

- Collaborated with a team of six to build a full-stack web app for buying and selling foreign currencies, integrating live exchange rate APIs and a secure user authentication system.
- Designed backend services with Node.js/Express and connected to a MongoDB database for transaction tracking.
- Developed an intuitive React-based frontend for real-time price display and order management.

Multi-Core Operating System | *C++*

Aug. 2025 - Dec. 2025

- Implemented a preemptive multi-core OS kernel with multi-threading.
- Built a virtual memory system with demand paging, page cache, swap space, and mmap.
- Implemented user level system calls for process/thread creation, file I/O, memory management, etc.

JPEB: 16-Bit Computer | *Rust, Haskell, C, Verilog, Python, Assembly*

April 2025

- Collaborated in a team of four to design and implement a custom 16-bit computer from scratch, including a custom ISA inspired by Dr. Bruce Jacob's RiSC-16.
- Developed a 6-stage pipelined processor, deployed it on an FPGA, and extended an existing compiler and assembler to support game development.
- Implemented the Chrome Dinosaur and Snake games using the custom 16-bit ISA; integrated graphics rendering into the emulator, increasing test coverage and validation efficiency by 40%.

Technical Skills / Additional Information

Languages: C/C++, Python, Julia, Java, HTML/CSS, JavaScript/TypeScript, Verilog, x86, ARM Assembly, LaTeX

Developer Tools: Git, Jupyter Notebook, Google Colab, VS Code, Docker, PyCharm, IntelliJ, Android Studio

Libraries: pandas, NumPy, Matplotlib, Scikit-Learn, TensorFlow, Keras